# CSE350: Digital Electronics & Pulse Techniques

Lecture 02: CSE251 Review

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# CSE251 Review

**Electronics:** *The method of controlling flow of charge carriers by application of electric field or other external forces.*

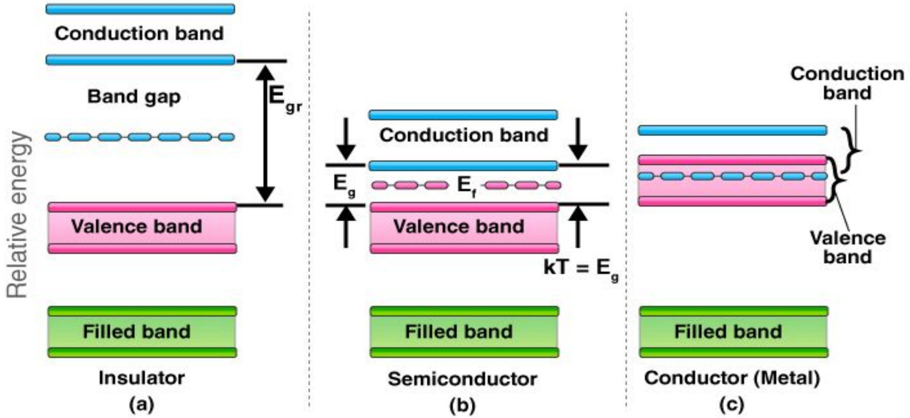
Similar sectors: Spintronics(Controls spin of electrons), Photonics(Controls photon emission or absorption), Twistronics, Valleytronics etc.

Classification of materials based on conductivity:

1. Metal (Good Conductor)
2. Semiconductor (Conducts at certain conditions)
3. Insulator (Bad Conductor)

**Band Theory:** *The energy of an electron in a large sample of any material is not a singular value, rather the different values form a band around the unique value that would've been if it were in a single atom.*

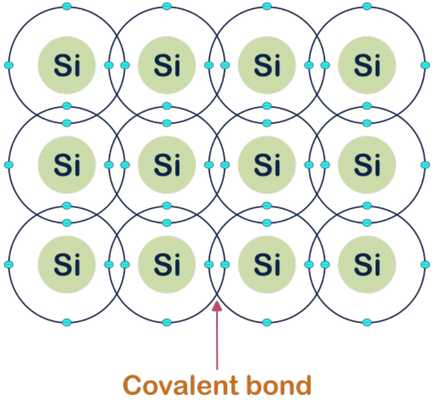
There are two types of bands we need to consider: Conduction & Valance band. For the 3 types of materials mentioned above, the energy bands have roughly the following form:




Order of bandgap : Insulato(>3 eV), Semiconductor(0-3 eV), Metal

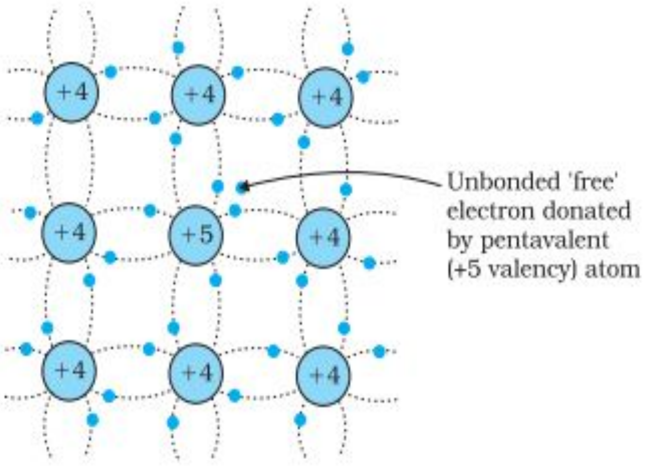
**Semiconductor:** Two types- i. Intrinsic semiconductor & ii. Extrinsic semiconductor.

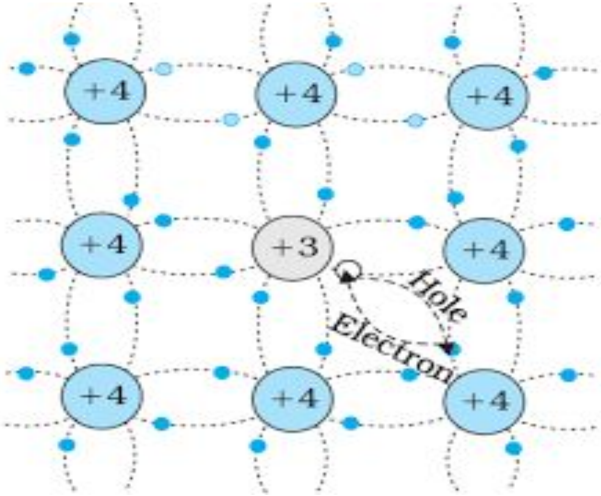
**Intrinsic semiconductors** are made of pure Si/Ge/GaAs or other semiconductor materials, containing no impurities. For instance, a Si-semiconductor has the following structure:




**Extrinsic semiconductors** are formed by adding Group-13/15 impurities to these intrinsic ones. This process is called doping and based on doping material it is either n-type(Group-15 doping) or p-type(Group-13 doping).

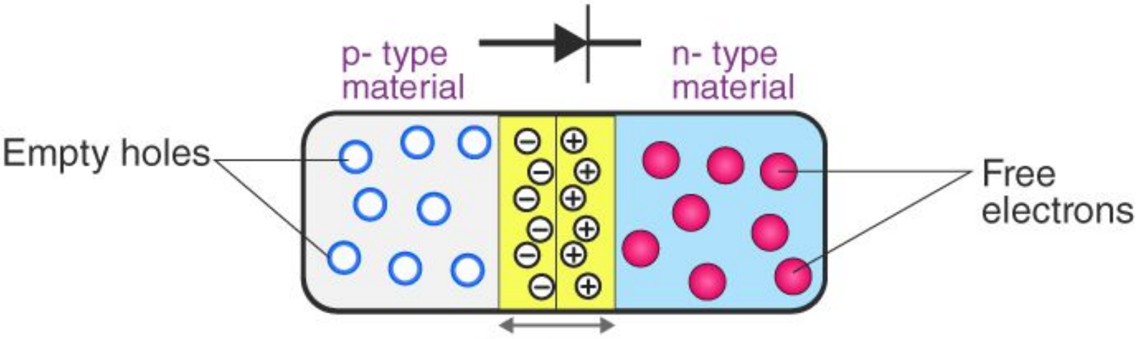
| n-type semiconductor | p-type semiconductor |
| --- | --- |



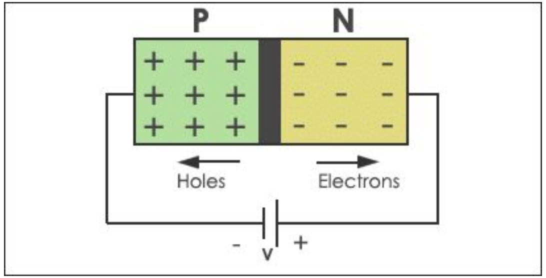

Majority carrier in an n-type semiconductor is electron & in p-type, hole (absence of electrons).

## p-n junction diode:



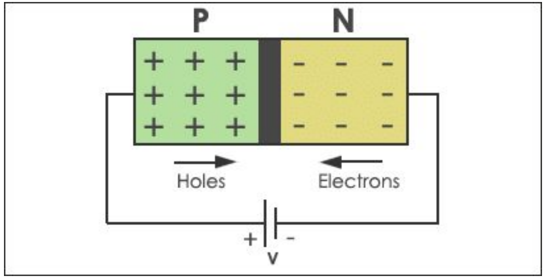

At the point of connection, opposite majority carriers from both type of semiconductors combine and form a depletion region (depleted of free charges). There exists negative ions in p-type part & positive ions in n-type part of the depletion region. These ions repulse further flow of majority carriers in either direction.

**Reverse Bias:**



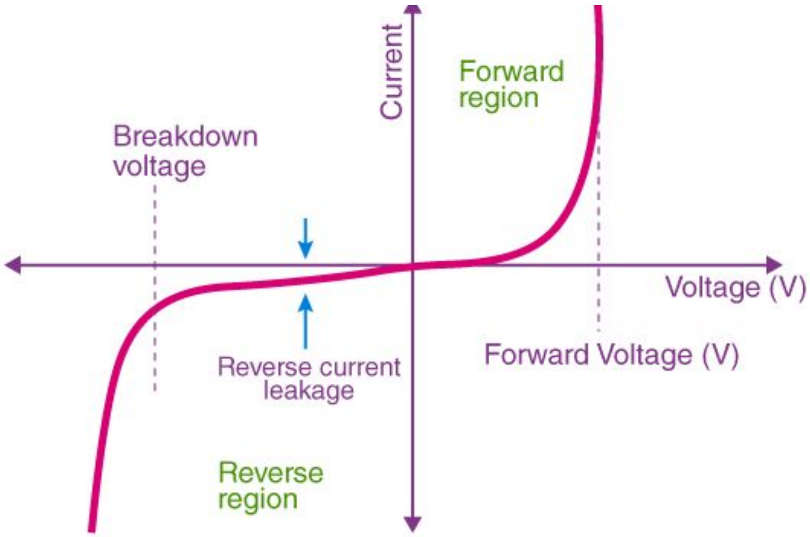

This biasing increases the depletion region further. Only a very low current due to minority carriers might flow (called Reverse Saturation Current).

**Forward Bias:**




It reduces the depletion region, and an exponential current results in forward direction.

## I-V Characteristics:



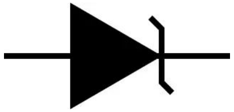

**Diode current,**

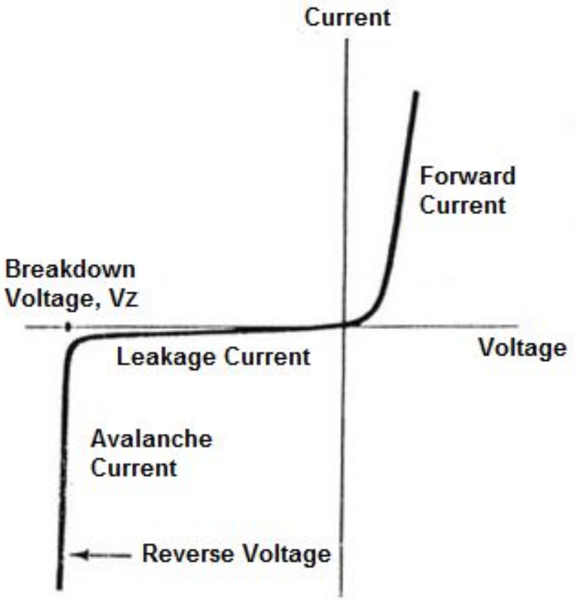
$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

Breakdown occurs, when applied reverse bias voltage is too high. While this is undesirable in a regular diode, Zener Diodes are operated in breakdown region, with a safer breakdown mechanism.

## Zener Diode I-V Characteristics:

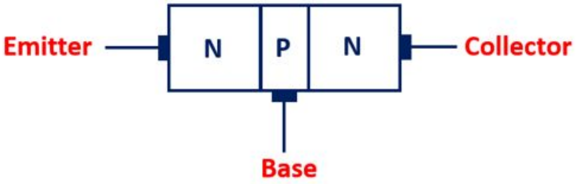
**Symbol:**

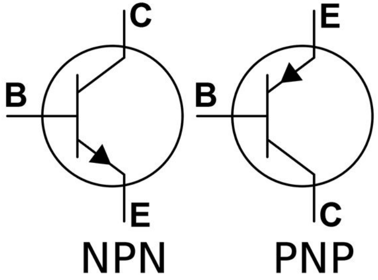





When reverse bias voltage reaches breakdown voltage, Zener diode becomes active and a constant voltage $V_z$ is dropped across the diode, in reverse polarity.

## BJT (Bipolar Junction Transistor):




## Modes of operation:

| B-E Junction | B-C Junction | Mode |
| --- | --- | --- |
| Forward | Reverse | Forward Active |
| Forward | Forward | Saturation |
| Reverse | Reverse | Cutoff |
| Reverse | Forward | Reverse Active |

$V_{BC}$

$V_{BE}$

## Reverse Active:

Emitter & Collector switch their roles

$$V_{BC} = 0.7V, \frac{I_E}{I_B} = \beta_R$$

Here, $\beta_R$ is reverse beta, which typically values between 0.1-0.3. Current flows from Emitter to Collector. $V_E >> V_B > V_C$

## Saturation:

$$V_{BE} = 0.8V, V_{CE} = 0.1 - 0.2V, V_{BC} = 0.6V - 0.7V$$

**Checking Condition:**

$$\frac{I_C}{I_B} = \beta_{forced} < \beta_{F(Forward)}$$

## Cutoff:

( $V_{BE} < 0, V_{BC} < 0$ )

$$I_C = I_E = I_B = 0$$

$V_{BE} < V_\gamma$ = Cut-in voltage/Turn-on voltage of Transistor = 0.5 V

## Forward Active:

$$V_{BE} > 0, V_{BC} < 0$$

$$V_{BE} = 0.7V, \frac{I_C}{I_B} = \beta_F$$

Here, $\beta_F$ is forward beta, which typically values 30, 50, 100 or more.

**Checking condition:** $V_{CE} > 0.2V$